Grade 10 – Test 7

Name: _____
Grade / Class: 10

ENGLISH (Language, Grammar, Literature, Writing)

1. **Rewrite the sentence by replacing the underlined phrase with a suitable idiom:**
   She gave up the plan because it was too risky.
2. **Identify the tense used in the sentence and justify your answer:**
   By the time the bell rang, the assembly had already ended.
3. **Combine the sentences using a relative clause:**
   The scientist made a discovery. The discovery changed medical history.
4. **Convert the sentence into indirect speech:**
   Sara said, "I will complete the assignment after dinner."
5. **Choose the correct word and rewrite the sentence:**
   Neither the manager nor the employees (is / are) satisfied with the decision.
6. **Explain the theme of perseverance using an example from real life.**
7. **Write a descriptive paragraph about a place that becomes completely silent at night.**
8. **Identify the type of sentence and rewrite it as an interrogative:**
   You finished the project successfully.
9. **Read the extract:**
   "The old bridge stood firm despite decades of storms and floods."
   State what the bridge symbolizes.
10. **Write a formal letter requesting permission to organize a school science exhibition.**

MATHEMATICS

1. **Factorize completely:**
   $4x^2 - 25y^2$
2. **Solve for x:**
   $3(x - 4) + 2 = 5x - 10$
3. **If the radius of a circle is increased by 50 percent, explain how the area changes.**
4. **Find the value of k if the points (2, k), (6, 10), and (10, 18) lie on a straight line.**
5. **Simplify:**
   $(3a^{-2} b^3) \times (4a^3 b^{-1})$
6. **A rectangular field has a perimeter of 96 meters. If its length is 28 meters, find its width.**

7. **Calculate the mean of the first ten natural odd numbers.**
8. **Solve the quadratic equation using any method:**
   $x^2 - 7x + 12 = 0$
9. **Find the volume of a cone with height 14 cm and base radius 7 cm.**
10. **If sin A = 5/13, where A is acute, find cos A.**

PHYSICS

1. **State Newton's Third Law of Motion and give one practical example.**
2. **Explain why passengers feel pushed backward when a bus suddenly starts moving.**
3. **Define electric potential difference and mention its SI unit.**
4. **A car travels 120 km in 2 hours. Calculate its average speed.**
5. **Distinguish between scalar and vector quantities with examples.**
6. **Describe how energy transformation occurs in a hydroelectric power station.**
7. **Why does a metal spoon feel colder than a wooden spoon at the same temperature?**
8. **Draw and label a ray diagram for reflection in a plane mirror.**
9. **Define momentum and write its formula.**
10. **Explain the role of fuses in electrical circuits.**

CHEMISTRY

1. **Define chemical equilibrium and state one condition required to achieve it.**
2. **Write the chemical formula of washing soda and state one of its uses.**
3. **Differentiate between exothermic and endothermic reactions.**
4. **Explain why rusting of iron is faster in coastal areas.**
5. **Balance the chemical equation:**
   $Al + O_2 \rightarrow Al_2O_3$
6. **Define pH scale and mention the pH value of a neutral solution.**
7. **What is meant by homologous series? Give one example.**
8. **State two properties of acids that distinguish them from bases.**
9. **Describe the process of fractional distillation of petroleum.**
10. **Why are noble gases considered chemically inert?**

BIOLOGY

1. **Define homeostasis and explain its importance in living organisms.**
2. **Describe the function of red blood cells in the human body.**

3. **Differentiate between asexual and sexual reproduction.**
4. **Explain how transpiration benefits plants.**
5. **Name the part of the brain responsible for balance and coordination.**
6. **Describe the process of photosynthesis with a balanced equation.**
7. **What is genetic variation and why is it important for survival?**
8. **State the role of enzymes in digestion.**
9. **Explain the difference between arteries and veins.**
10. **Define ecosystem and list its main components.**

COMPUTER SCIENCE / ICT

1. **Define algorithm and explain why it is important in programming.**
2. **Differentiate between system software and application software.**
3. **What is a flowchart? Name any two symbols used in it.**
4. **Explain the concept of data security with one example.**
5. **Write two advantages of using cloud storage.**
6. **What is a database? State one use of databases in daily life.**
7. **Explain the purpose of an operating system.**
8. **Define input device and name two examples.**
9. **What is the difference between RAM and ROM?**
10. **Explain how the internet has changed communication methods.**

PAKISTAN STUDIES / SOCIAL STUDIES (Global Standard)

1. **Explain the importance of natural resources in national development.**
2. **Describe the causes of population growth and its effects.**
3. **What is climate change? Mention two human activities contributing to it.**
4. **Explain the role of trade in strengthening a country's economy.**
5. **Describe the functions of local government.**
6. **What is meant by cultural diversity? Give one example.**
7. **Explain why water conservation is necessary.**
8. **Define globalization and mention one advantage.**
9. **Describe the impact of deforestation on the environment.**
10. **Explain the importance of education in social development.**